Nuclear fission barrier heights in covariant energy density functional theory

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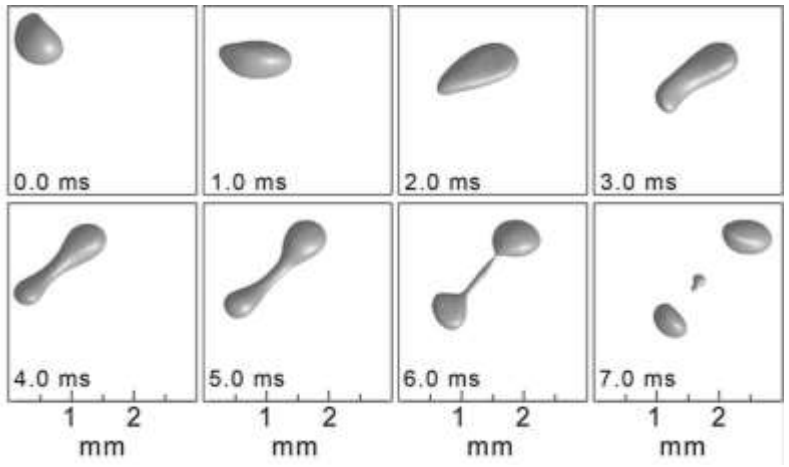


Seminar 2026.9.4

Different droplets breakup



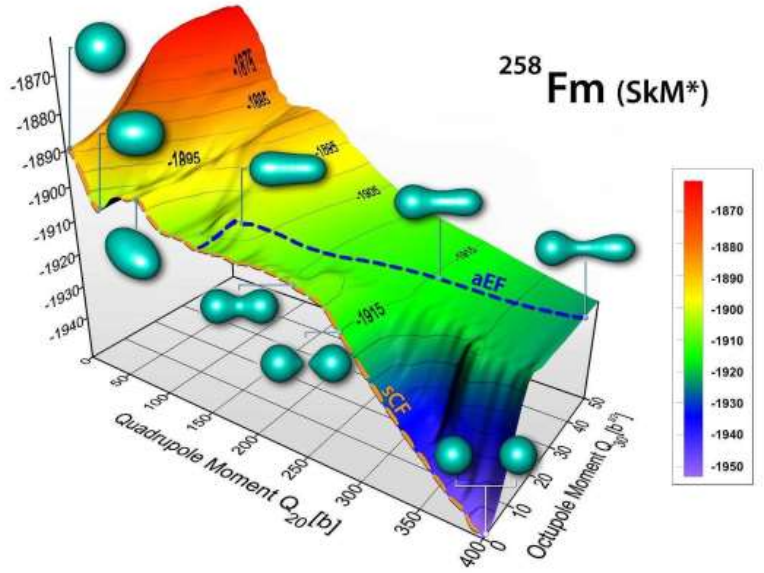
Macroscopic droplet breakup



M. Karimi and R. Andersson, Chem. Eng. J. 380, 122502 (2020).

- Mechanism: electromagnetic interaction.
- Macroscopic (hydrodynamics), not spontaneous.
- Time scale: ~ms.

Microscopic nuclear fission



A. Staszczak et al. Phys. Rev. C 80, 014309 (2009)

- Mechanism: strong interaction, electromagnetic interaction.
- Microscopic (shell effect, pairing effect), spontaneous (quantum tunneling).
- Time scale: ~ 10^{-20} s.

- In the liquid-drop model, the binding energy of an atomic nucleus ^{A}Z at spherical shape is

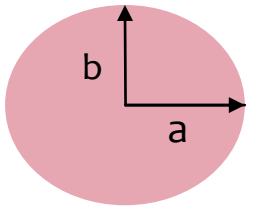
$$E_{\rm LDM} = a_v A + a_s A^{2/3} + a_c \frac{Z^2}{A^{1/3}} + a_{\rm sym} \frac{(N-Z)^2}{A} + \delta(A).$$

- For the deformed nuclei, the surface energy and Coulomb energy are

N. Bohr and J. A. Wheeler, Physical Review 56, 426 (1939)

$$E_{\rm surf}(\epsilon) \approx E_{\rm surf}(0)\left(1 + \frac{2}{5}\epsilon^2 - \frac{52}{105}\epsilon^3 + \ldots\right),$$

$$E_{\rm Coul}(\epsilon) \approx E_{\rm Coul}(0)\left(1 - \frac{1}{5}\epsilon^2 + \frac{4}{21}\epsilon^3 + \ldots\right).$$



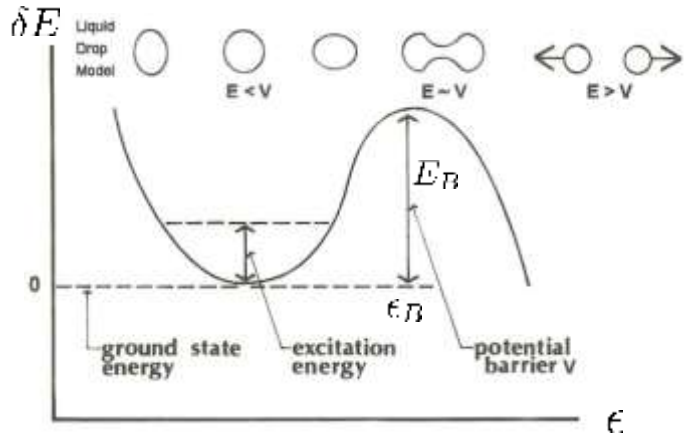
$$a = R(1+\epsilon),$$
$$b = R(1+\epsilon)^{-1/2},$$
$$ab^2 = R^3 = \text{constant}.$$

- The energy difference is

$$\Delta E = E_{\rm surf}^{(0)}\left\{\frac{2}{5}(1-x)\epsilon^2 - \frac{4}{105}(13-10x)\epsilon^3 + \cdots\right\}, \quad x \equiv \frac{E_{\rm Coul}(0)}{2E_{\rm surf}(0)} \approx 0.02\, Z^2/A$$

With the fission barrier

$$E_B = \frac{98}{15}\frac{(1-x)^3}{(13-10x)^2}E_s^{(0)}, \qquad \epsilon_B = \frac{7(1-x)}{13-10x}$$

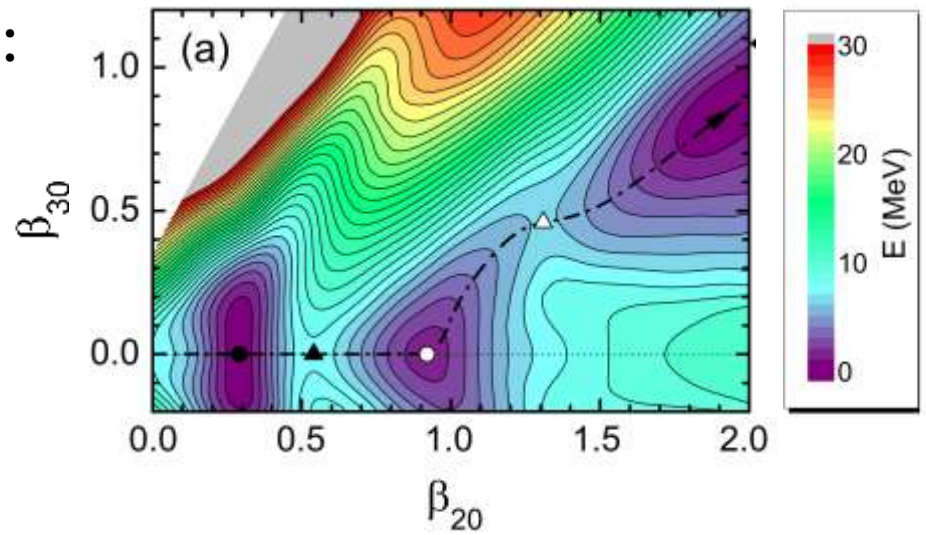


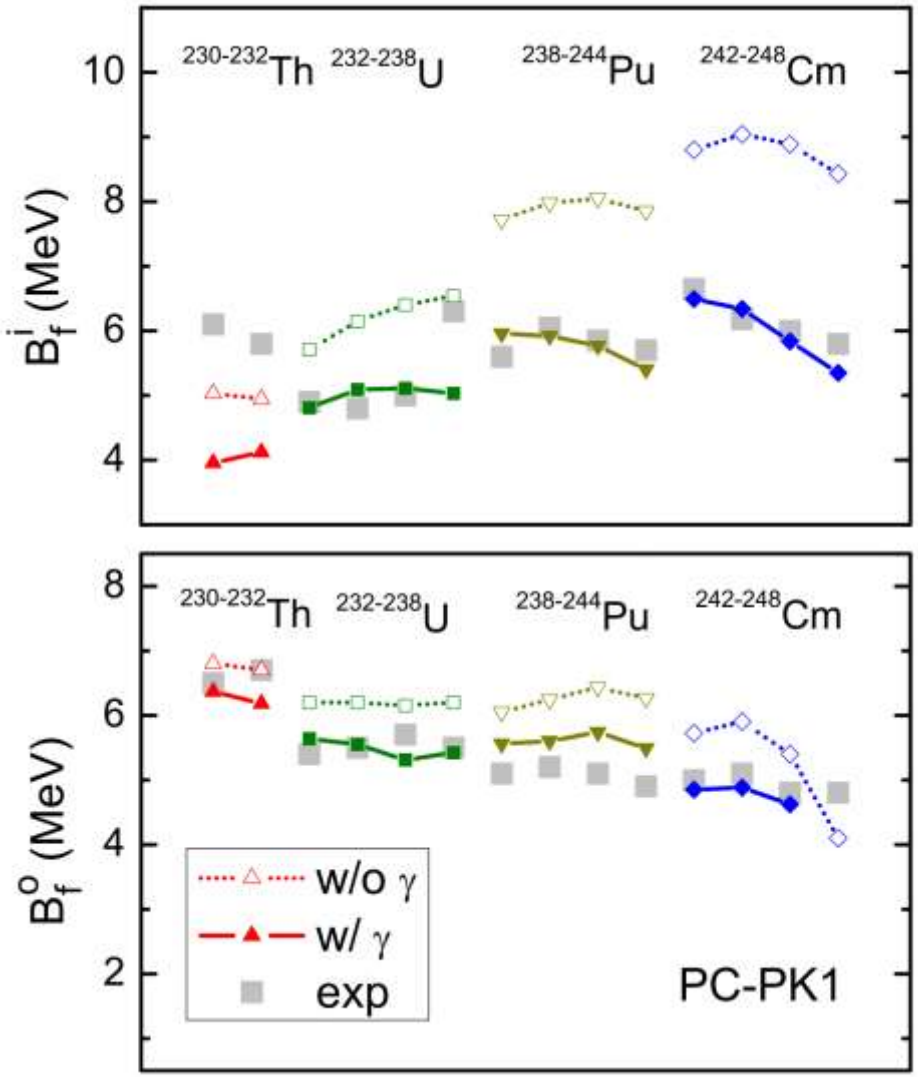
- Shell effect contribute the fission barrier at Z>104.
- The microscopic picture, including single-particle behavior, needs to be clearer.
- Nuclear density functional theory is one of the most successful microscopic frameworks for describing nuclear fission.

K.-H. Schmidt and B. Jurado, Reports on Progress in Physics 81,106301 (2018)

M. Bender et al., J. Phys. G 47, 113002 (2020)

^{240}Pu :





B.-N. Lu et al., EPJ Web of Conferences 38, 05003 (2012)

Relativistic point-coupling Density Functional Theory

- The expectation of Hamiltonian becomes :

$$E_{\mathrm{DF}}\left[\boldsymbol{\tau}, \rho_S, j_i{}^{\mu}, A_{\mu}\right] = \int d^3r \left(\mathcal{E}^{\mathrm{kin}}(\boldsymbol{r}) + \mathcal{E}^{\mathrm{int}}(\boldsymbol{r}) + \mathcal{E}^{\mathrm{em}}(\boldsymbol{r})\right)$$

- The interaction part with low-energy constants $\mathbf{C} = \{\alpha_S, \beta_S, \gamma_S, \delta_S, \alpha_V, \gamma_V, \delta_V, \alpha_{TV}, \delta_{TV}\}$:

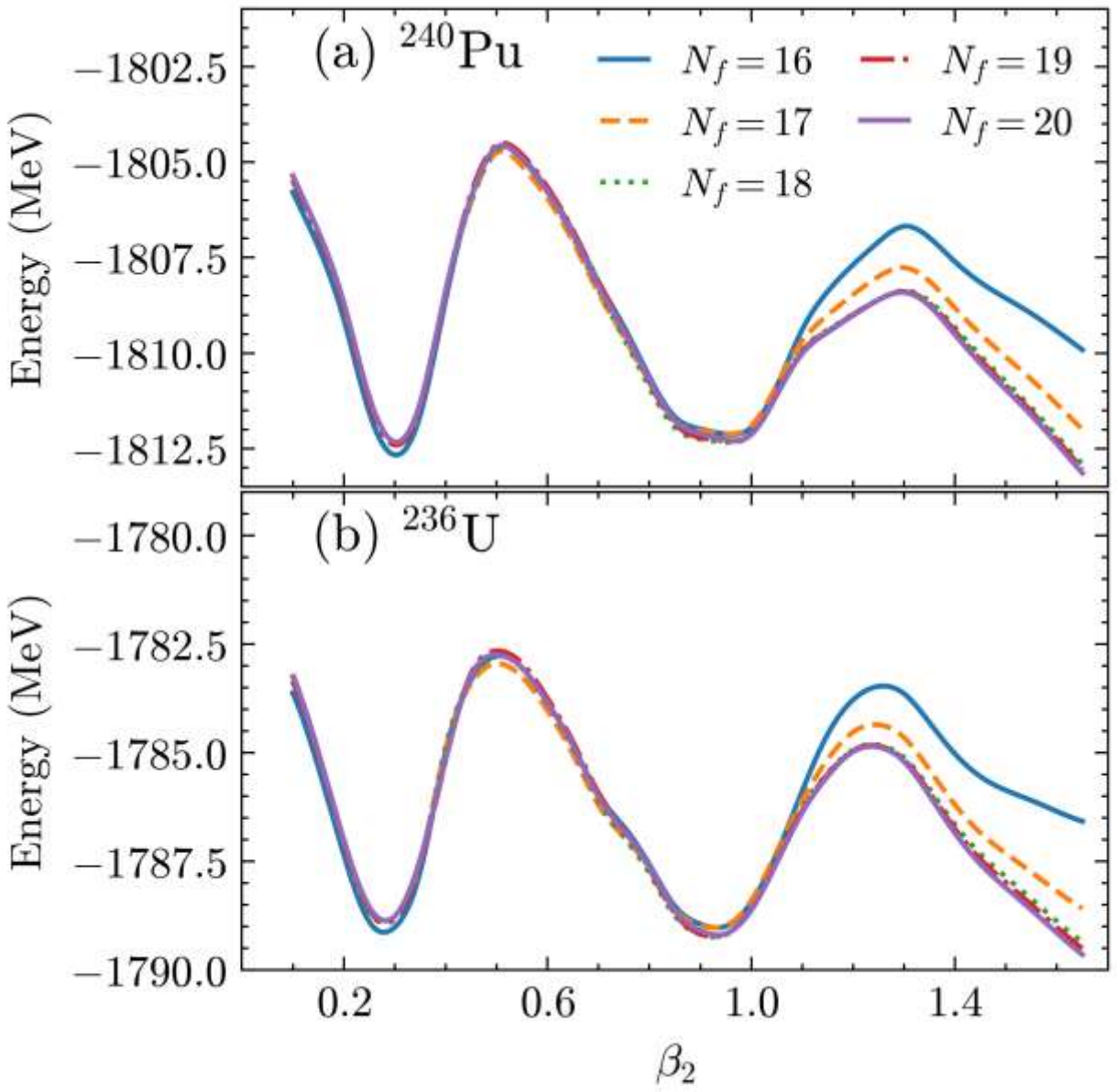
$$\begin{aligned}\mathcal{E}^{\mathrm{int}}(\boldsymbol{r}) =& \frac{\alpha_S}{2}\rho_S^2 + \frac{\beta_S}{3}\rho_S^3 + \frac{\gamma_S}{4}\rho_S^4 + \frac{\delta_S}{2}\rho_S\Delta\rho_S \\ &+ \frac{\alpha_V}{2}j_\mu j^\mu + \frac{\gamma_V}{4}\left(j_\mu j^\mu\right)^2 + \frac{\delta_V}{2}j_\mu \Delta j^\mu \\ &+ \frac{\alpha_{TV}}{2}\vec{j}_{TV}^{\,\mu}\cdot\left(\vec{j}_{TV}\right)_\mu + \frac{\delta_{TV}}{2}\vec{j}_{TV}^{\,\mu}\cdot\Delta\left(\vec{j}_{TV}\right)_\mu .\end{aligned}$$

- The local densities $\rho_S(\boldsymbol{r})$ and currents $j_V^{\mu}(\boldsymbol{r})$, $\vec{j}_{TV}^{\,\mu}(\boldsymbol{r})$ are :

$$\rho_S(\mathbf{r}) = \sum_{k=1}^{A<\epsilon_F} \bar{\psi}_k(\mathbf{r})\psi_k(\mathbf{r}), \qquad j^{\mu}(\mathbf{r}) = \sum_{k=1}^{A<\epsilon_F} \bar{\psi}_k(\mathbf{r})\gamma^{\mu}\psi_k(\mathbf{r}),$$

$$j_{TV}^{\mu}(\mathbf{r}) = \sum_{k=1}^{A<\epsilon_F} \bar{\psi}_k(\mathbf{r})\vec{\tau}\,\gamma^{\mu}\psi_k(\mathbf{r}).$$

- The total energy : $E_{\mathrm{tot}} = E_{\mathrm{DF}} + E_{\mathrm{pair}}^{\tau} + E_{\mathrm{cm}}^{\mathrm{mic}} + E_{\mathrm{rot}} + E_{\mathrm{ext}}$



- The potential of the axially deformed harmonic oscillator basis is

$$V_{\rm osc}(z, r_\perp) = \frac{1}{2} m\omega_z^2 z^2 + \frac{1}{2} m\omega_\perp^2 r_\perp^2 .$$

with

$$\omega_z = \omega_0 \exp(-\sqrt{\frac{5}{4\pi}}\beta_{\rm HO}),$$

$$\omega_\perp = \omega_0 \exp(\frac{1}{2}\sqrt{\frac{5}{4\pi}}\beta_{\rm HO}).$$

- $\beta_{\rm HO}$ is chosen as $\beta_{\rm HO} = \beta_2$ for $\beta_2 < 0.4$ and $\beta_{\rm HO} = \beta_2/2$ for $\beta_2 \geq 0.4$.

- The initial woods-Saxon potential is

$$V(r, \theta) = \frac{V_0}{1 + \exp\left[\dfrac{r - Rf(\theta)}{a}\right]} \quad , \text{ with}$$

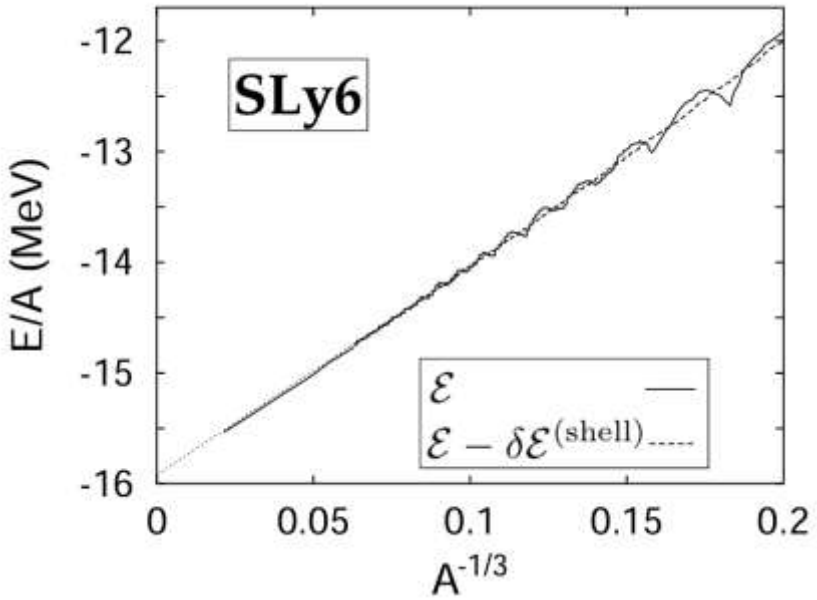
$$f(\theta) = 1 + \frac{3}{4}\left(\beta_2^{\rm W} Y_{20}(\theta) + \beta_3^{\rm W} Y_{30}(\theta)\right)$$

- $\beta_3^{\rm W}$ is chosen as the corresponding PES valley for $\beta_2 < 1.4$ and $\beta_3^{\rm W} = 0.5$ for $\beta_2 \geq 1.4$.

Extraction of the surface energy coefficient

- From LDM :

$$\frac{E}{A} = a_v + a_s A^{-1/3} + a_{\rm sym} I^2 + \delta\mathcal{E}^{(\rm shell)}$$



P.-G. Reinhard et al., PRC 73, 014309 (2006).

<2% (I=0)

- From semi-infinite nuclear matter (SINM) :

$$\sigma(I) = \frac{E_{\rm SINM} - E_{\rm bulk}}{S} + \mu\,\frac{N_{\rm bulk} - N_{\rm SINM}}{S}$$

$$a_s(I) = 4\pi\, r_0^2(I)\,\sigma(I)$$

HF : construct the EDF of SINM

↓

Minimize the EDF

↓

Obtain the Dirac equation for single particle

↓

Obtain the density

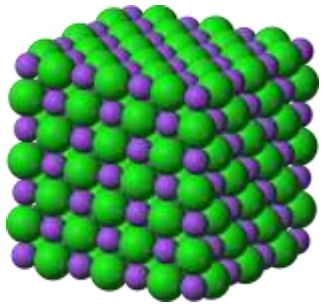
↑ (loop back)

Thomas-Fermi (TF) approximation

~3% (I=0)

Nuclear matter properties

L



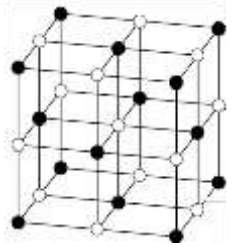
r space

$$\left[\boldsymbol{\alpha}\cdot\mathbf{k}+\beta\underbrace{M_\tau^*+\Sigma_\tau^0}\right]\psi_{\mathbf{k}}^\tau=\epsilon_{\mathbf{k}}^\tau\psi_{\mathbf{k}}^\tau$$

constant

$$\begin{aligned}M_\tau^* &= M_\tau+\alpha_S\rho_S+\beta_S\rho_S^2+\gamma_S\rho_S^3\\ \Sigma_\tau^0 &= \alpha_V\rho_V+\gamma_V\rho_V^3+\alpha_{TV}\tau_3\rho_{TV}\end{aligned}$$

$$\begin{cases}\psi_{\mathbf{k}}^\tau(\mathbf{r})=\dfrac{1}{\sqrt{V}}u_{\mathbf{k}}^\tau e^{i\mathbf{k}\cdot\mathbf{r}}\\ \epsilon_{\mathbf{k}}^\tau=E_\tau^*(k)+\Sigma_\tau^0\end{cases}\longrightarrow\quad k_i=\frac{2\pi n_i}{L},\qquad n_i=0,\pm1,\pm2,\ldots$$



$\frac{2\pi}{L}$

p space

$$\rho_S^{(\tau)}=\sum_{\mathbf{k}\lambda}^{\text{occ}}\bar{\psi}_{\mathbf{k}\lambda}^{(\tau)}\psi_{\mathbf{k}\lambda}^{(\tau)}=\frac{1}{V}\sum_{\mathbf{k}\lambda}^{\text{occ}}\bar{u}_{\mathbf{k}\lambda}^{(\tau)}u_{\mathbf{k}\lambda}^{(\tau)}$$

$$\xrightarrow{V\to\infty}\sum_\lambda\int_{|\mathbf{k}|\le k_{\tau F}}\frac{d^3k}{(2\pi)^3}\bar{u}_{\mathbf{k}\lambda}^{(\tau)}u_{\mathbf{k}\lambda}^{(\tau)}=\frac{2}{(2\pi)^3}\int_0^{k_{\tau F}}\frac{M_\tau^*}{E_\tau^*(\mathbf{k})}d^3k,$$

$$\rho_V^{(\tau)}=\sum_{\mathbf{k}\lambda}^{\text{occ}}\psi_{\mathbf{k}\lambda}^{(\tau)\dagger}\psi_{\mathbf{k}\lambda}^{(\tau)}=\frac{1}{V}\sum_{\mathbf{k}\lambda}^{\text{occ}}u_{\mathbf{k}\lambda}^{(\tau)\dagger}u_{\mathbf{k}\lambda}^{(\tau)}$$

$$\xrightarrow{V\to\infty}\sum_\lambda\int_{|\mathbf{k}|\le k_{\tau F}}\frac{d^3k}{(2\pi)^3}u_{\mathbf{k}\lambda}^{(\tau)\dagger}u_{\mathbf{k}\lambda}^{(\tau)}=\frac{k_{\tau F}^3}{3\pi^2}$$

In the k-space, the total volume is $\frac{4}{3}\pi k_F^3$, then the number of particles is $\frac{V}{3\pi^2}k_F^3$. The particle number density is $\rho=\frac{N}{V}=\frac{1}{3\pi^2}k_F^3$.

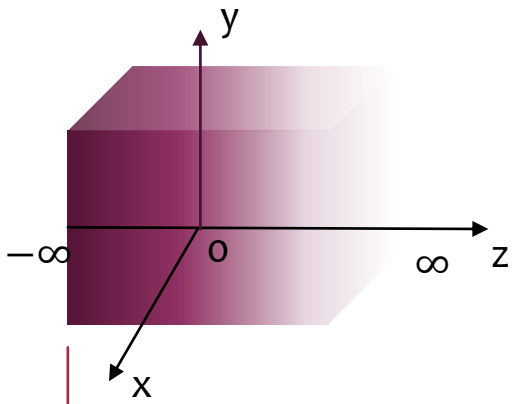
$$\mathcal{E}(\rho_V,I)=\sum_\tau\left[\frac{3}{4}\left(\rho_V^{(\tau)}E_{\tau F}^*-\rho_S^{(\tau)}M_\tau^*\right)+\rho_V^{(\tau)}M_\tau\right]+\frac{\alpha_S}{2}\rho_S^2+\frac{\beta_S}{3}\rho_S^3+\frac{\gamma_S}{4}\rho_S^4+\frac{\alpha_V}{2}\rho_V^2+\frac{\gamma_V}{4}\rho_V^4+\frac{\alpha_{TV}}{2}\rho_{TV}^2$$

with $\rho_{TV}=\rho_V^{(n)}-\rho_V^{(p)}=I\rho_V$

Extraction of the surface energy coefficient from SINM

$$E_{\rm bulk} = \int \mathcal{E}(\rho_0)\, d^3r = S\int \mathcal{E}(\rho_0)\, dz,$$

$$E_{\rm SINM} = \int \left[\mathcal{E}(\rho) + C(\rho)\,(\nabla\rho)^2\right] d^3r = S\int \left[\mathcal{E}(\rho) + C(\rho)\,\rho'^2\right] dz$$



- $\sigma(I) = \dfrac{E_{\rm SINM} - E_{\rm bulk}}{S} + \mu\,\dfrac{N_{\rm bulk} - N_{\rm SINM}}{S}$ with $\mu = \left.\dfrac{\partial\mathcal{E}}{\partial\rho}\right|_{\rho_0}$

  → $\sigma = \displaystyle\int_{-\infty}^{\infty}\left[C(\rho)\,\rho'^2 + w(\rho)\right] dz,$ with $w(\rho) \equiv \mathcal{E}(\rho) - \mu\rho$

$$P(\rho_0) = \mu\rho_0 - \mathcal{E}(\rho_0) = 0 \rightarrow \mu = \frac{\mathcal{E}(\rho_0)}{\rho_0}$$

M. Centelles et al., Nuclear Physics A635, 193 (1998)

- We make variation of the surface tension $\delta\sigma = 0$. Under the TF approximation, all the densities can be determined uniquely from ρ_V, thus, we make variation and yield

$$C'(\rho)\,\rho'^2 + 2C(\rho)\,\rho'' = \frac{d\mathcal{E}}{d\rho} - \mu \quad\Longleftrightarrow\quad \frac{d}{dz}\left[C(\rho)\,\rho'^2\right] = \frac{d}{dz}\left[\mathcal{E}(\rho) - \mu\rho\right] = \frac{dw}{dz}$$

boundary condition → $C(\rho)\,\rho'^2 = w(\rho)$

- Using $\rho' = \sqrt{w/C}$, we can obtain : $\sigma(I) = 2\displaystyle\int_0^{\rho_0(I)} \sqrt{C(\rho)\,w(\rho)}\, d\rho.$

- From CDFT, the gradient part of the total energy is

$$E_{\rm grad} = \frac{1}{2}\int \left[\delta_S\,\rho_S\,\Delta\rho_S + \delta_V\,\rho_V\,\Delta\rho_V + \delta_{TV}\,\vec{\rho}_{TV}\cdot\Delta\vec{\rho}_{TV}\right] d^3r.$$

$$\longleftrightarrow \quad E_{\rm grad} = -\frac{S}{2}\int \left[\delta_V + \delta_S\left(\frac{d\rho_S}{d\rho_V}\right)^2 + \delta_{TV}\,I^2\right](\rho_V')^2\,dz$$

- Comparison with the gradient term $C(\rho)\,\rho'^2$, one can obtain

$$C(\rho) = -\frac{1}{2}\left[\delta_V + \delta_S\left(\frac{d\rho_S}{d\rho_V}\right)^2 + \delta_{TV}\,I^2\right]$$

And finally,

$$\sigma(I) = 2\int_0^{\rho_0(I)} \sqrt{-\frac{1}{2}\left[\delta_V + \delta_S\left(\frac{d\rho_S}{d\rho_V}\right)^2 + \delta_{TV}I^2\right]\left[\mathcal{E}(\rho_V, I) - \mu_n\rho_V^{(n)} - \mu_p\rho_V^{(p)}\right]}\,d\rho_V$$

- We define

$$a_{\rm surf} \equiv\ a_s(0) = 4\pi\,r_0^2(0)\,\sigma(0),$$
$$a_s(I) = a_{\rm surf} + a_{\rm ssym}\,I^2 + \mathcal{O}(I^4)$$

Approximation used:

1. The kinetic energy density $\tau(\mathbf{r})$ can be expressed by ρ_V and ρ_S ;
2. The relation between ρ_V and ρ_S is induced from TF approx.

	$a_{\rm surf}^{\rm (MTF)}$	$a_{\rm surf}^{\rm (ETF)}$	$a_{\rm surf}^{\rm (HF)}$	$a_{\rm ssym}^{\rm (HF)}$	$C_0^{\rho\nabla\cdot J}$
SLy5s1	18.00	17.15	17.55	−48.09	−86.61
SLy5s2	18.20	17.34	17.74	−48.21	−85.71
SLy5s3	18.40	17.53	17.93	−48.56	−84.65
SLy5s4	18.60	17.73	18.12	−49.01	−83.50
SLy5s5	18.80	17.92	18.31	−49.73	−82.31
SLy5s6	19.00	18.11	18.50	−50.53	−81.08
SLy5s7	19.20	18.31	18.70	−51.58	−79.82
SLy5s8	19.40	18.50	18.89	−52.70	−78.54

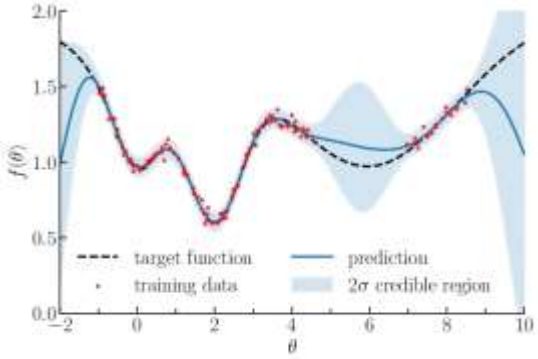
W. Ryssens et al., PRC 99, 044315 (2019).

- Scalar Gaussian Process (GP) :

$$O(\mathbf{C}) \in \{E_{\mathrm{g.s.}}, r_p, E_A, E_B, E_{\mathrm{Coul}}, x, a_{\mathrm{surf}}, a_{\mathrm{ssym}}, \rho_0, E/A, E_{\mathrm{sym}}, L, K, M^*/M\}$$

$$O(\mathbf{C}) \sim \mathcal{GP}\left(m(\mathbf{C}), k(\mathbf{C}, \mathbf{C}')\right)$$

Predictions : $O_*|\mathcal{D}, \mathbf{C}_* \sim \mathcal{N}(\mu_*, \sigma_*^2)$



- Principal Component Analysis (PCA)-GP:

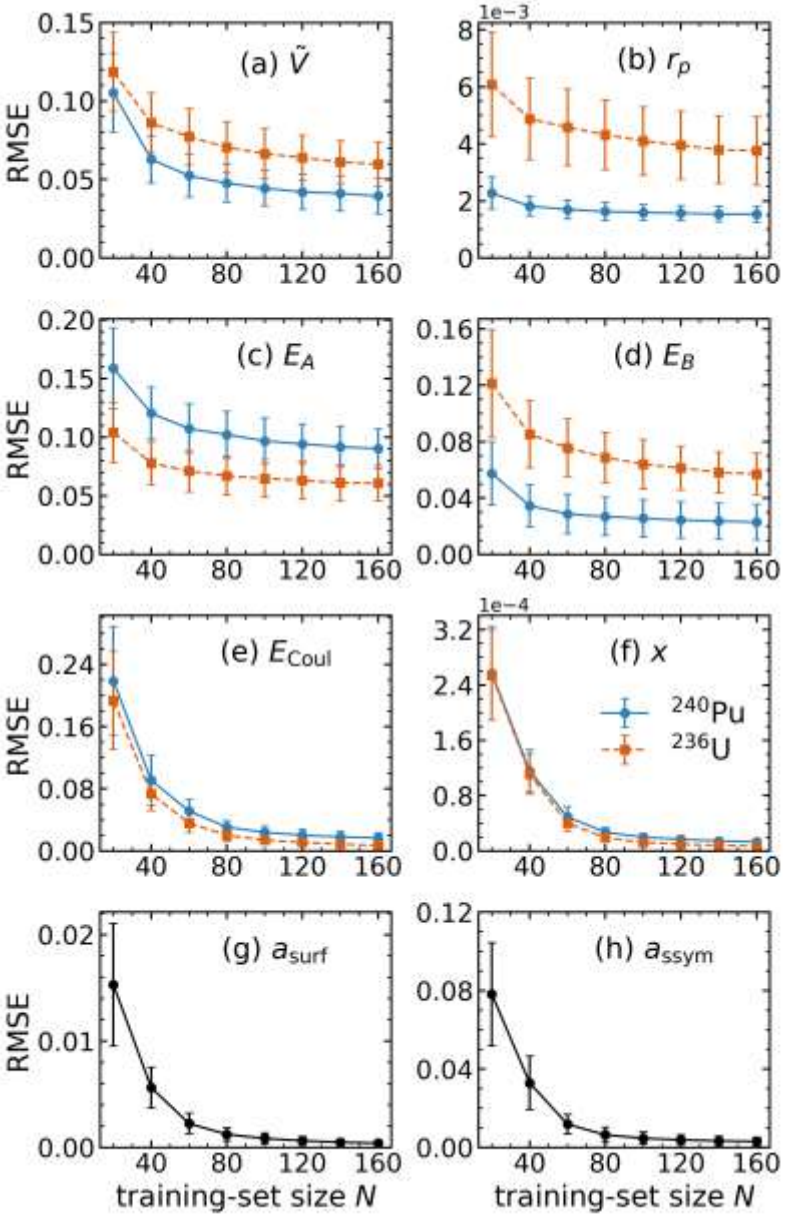
$$\mathbf{O}(\mathbf{C}) \in \{V(\beta_2),\ \beta_3(\beta_2)\}$$

$$\bar{V}(\beta_2^{(j)}) = \frac{1}{N}\sum_{i=1}^{N} V(\beta_2^{(j)}; \mathbf{C}_i), \quad Y_{ij} = V(\beta_2^{(j)}; \mathbf{C}_i) - \bar{V}(\beta_2^{(j)})$$

$$\mathbf{S} = \mathbf{Y}^\top \mathbf{Y}/(N-1) \xrightarrow{\text{eigenvalues}} \left\{ \lambda_c, \ \{\boldsymbol{\phi}_c\}_{c=1}^{M} \right. \longrightarrow c_k = \Big(\sum_{c=1}^{k} \lambda_c\Big)/\Big(\sum_{c=1}^{M} \lambda_c\Big)$$

$$V(\boldsymbol{\beta}_2; \mathbf{C}) \simeq \bar{V}(\boldsymbol{\beta}_2) + \sum_{c=1}^{k} z_c(\mathbf{C})\, \phi_c(\boldsymbol{\beta}_2) \ \text{ with } \ z_c(\mathbf{C}) \sim \mathcal{GP}$$

we choose $k = 6$ with $c_k > 99.9\%$



parameters	value	uncorr. error [%]	Train. range [%]
α_S	$-3.96291 \cdot 10^{-4}$	0.83	0.5
β_S	$8.6653 \cdot 10^{-11}$	4.7	2
γ_S	$-3.80724 \cdot 10^{-17}$	14	4
δ_S	$-1.09108 \cdot 10^{-10}$	23	20
α_V	$2.69040 \cdot 10^{-4}$	1.2	0.8
γ_V	$-3.64219 \cdot 10^{-18}$	54	30
δ_V	$-4.32619 \cdot 10^{-10}$	76	20
α_{TV}	$2.95018 \cdot 10^{-5}$	11	10
δ_{TV}	$-4.11112 \cdot 10^{-10}$	1700	150

P. W. Zhao et al., Phys. Rev. C 82, 054319 (2010)

- The ranges for sampling of the parameters are chosen according to the uncorrelated tolerance of parameters with $\chi^2 \leq \chi^2_{\min} + 1$;

  T. Bürvenich *et al.*, Phys. Rev. C 65, 044308 (2002)

- 200 training N_t using Latin hypercube sampling ;

- 6 million samples using quasi Monte-Carlo (MC) sampling with a uniform distribution.

- The posterior distribution of certain observable $O^{\rm pred}$ is

$$p(O^{\rm pred}|\mathcal{D},\mathbf{Q}^{\rm ref}) = \int p(O^{\rm pred}|\mathcal{D},\mathbf{C})\, p(\mathbf{C}|\mathcal{D},\mathbf{Q}^{\rm ref})\, d\mathbf{C} = \int p(O^{\rm pred}|\mathcal{D},\mathbf{C})\, \frac{\pi(\mathbf{C})\, p(\mathbf{Q}^{\rm ref}|\mathcal{D},\mathbf{C})}{p(\mathcal{D},\mathbf{Q}^{\rm ref})}\, d\mathbf{C}$$

$$p(O^{\rm given}|\mathcal{D},\mathbf{C}) = \int p(O^{\rm given}|O)\, p(O|\mathcal{D},\mathbf{C})\, dO$$

$$p(O^{\rm given}|O) = \mathcal{N}(O,\ \sigma^2_{\rm sys})$$

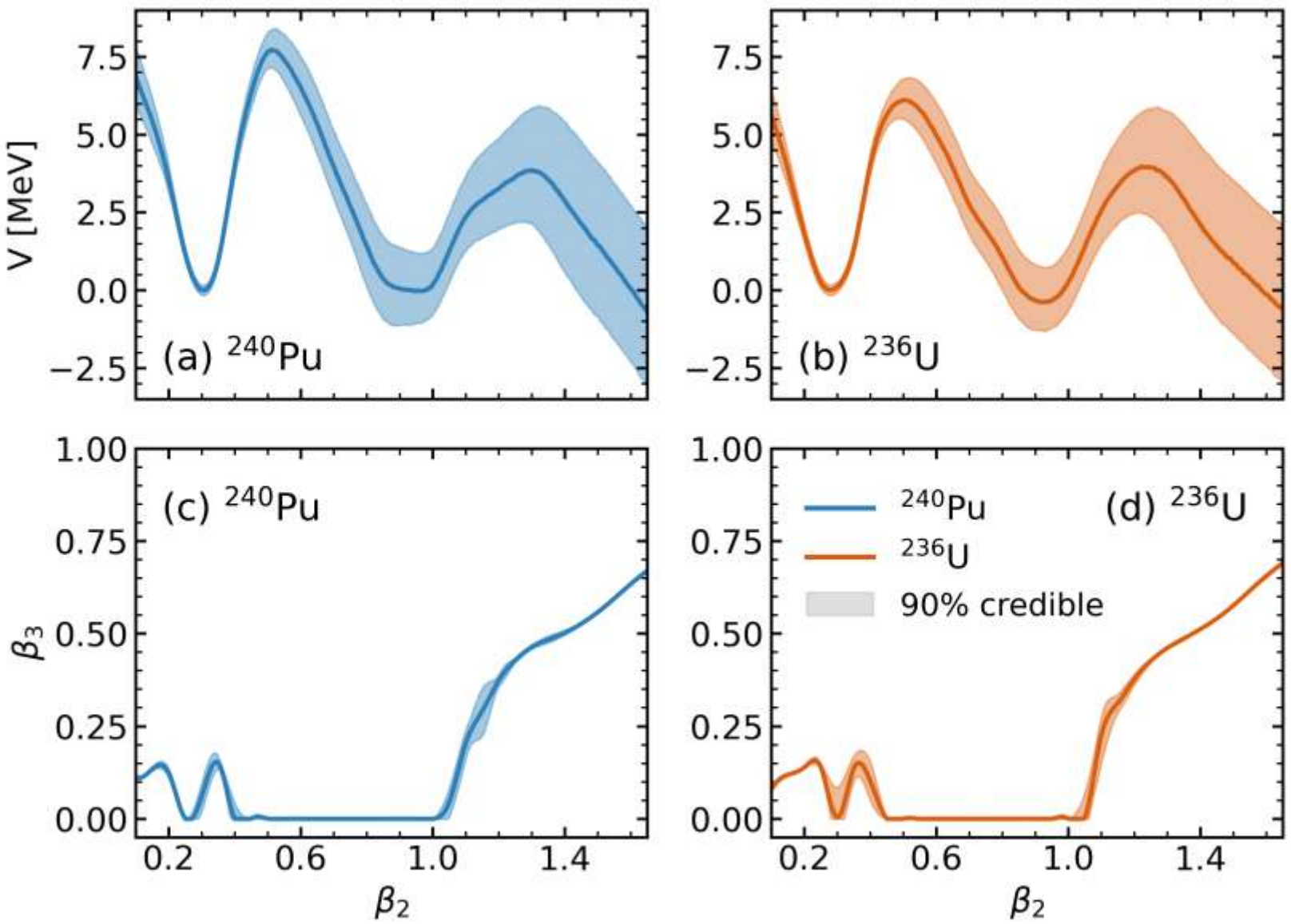
$$p(O|\mathcal{D},\mathbf{C}) = \mathcal{N}\big(\boldsymbol{\mu}_{\rm GP}(\mathbf{C}),\ \Sigma_{\rm GP}(\mathbf{C})\big)$$

$$p(\mathbf{Q}^{\rm ref}|\mathbf{Q}) = \mathcal{N}(\mathbf{Q},\ \Sigma_{\rm tol})$$

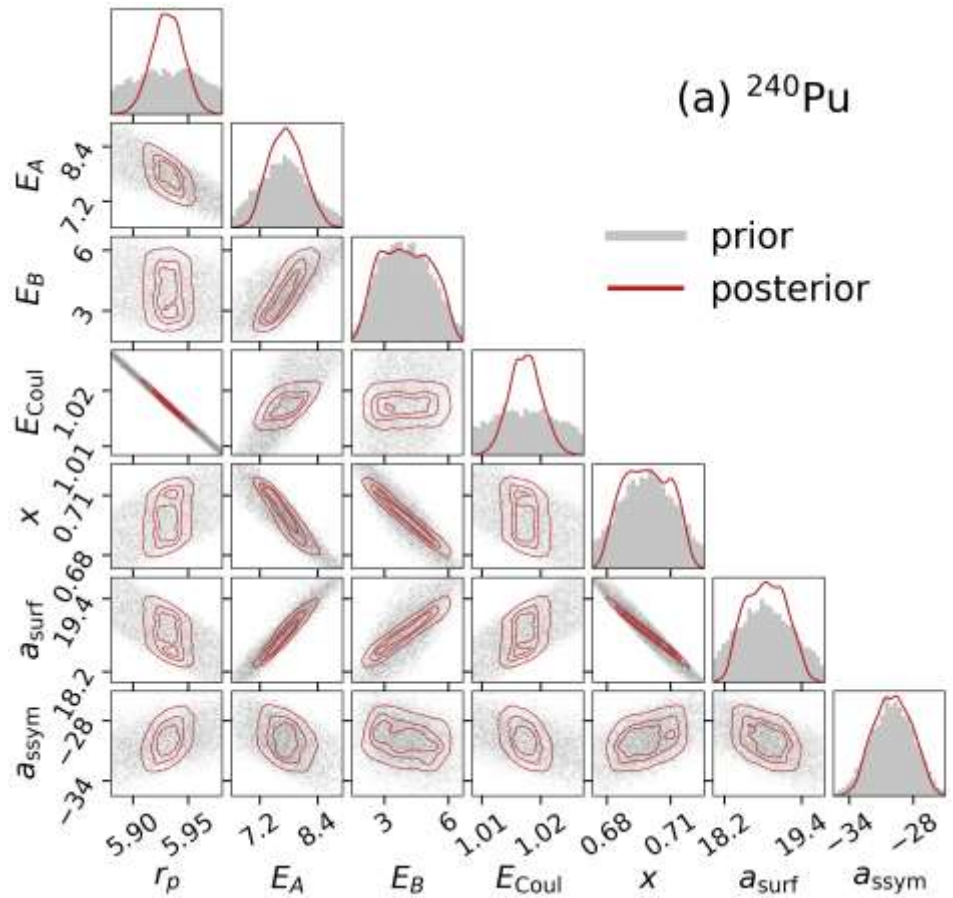
- Two sets of reference values are used : $\mathbf{Q}_1^{\rm ref} = \{E_{\rm g.s.}, r_p\}$, $\mathbf{Q}_2^{\rm ref} = \{\rho_0, E/A, E_{\rm sym}, L, K_\infty, m^*/m\}$.
  The tolerance for $\mathbf{Q}_1^{\rm ref}$ is $\sigma_{\rm tol}^1 = \{1\,{\rm MeV}, 0.02\,{\rm fm}\}$.

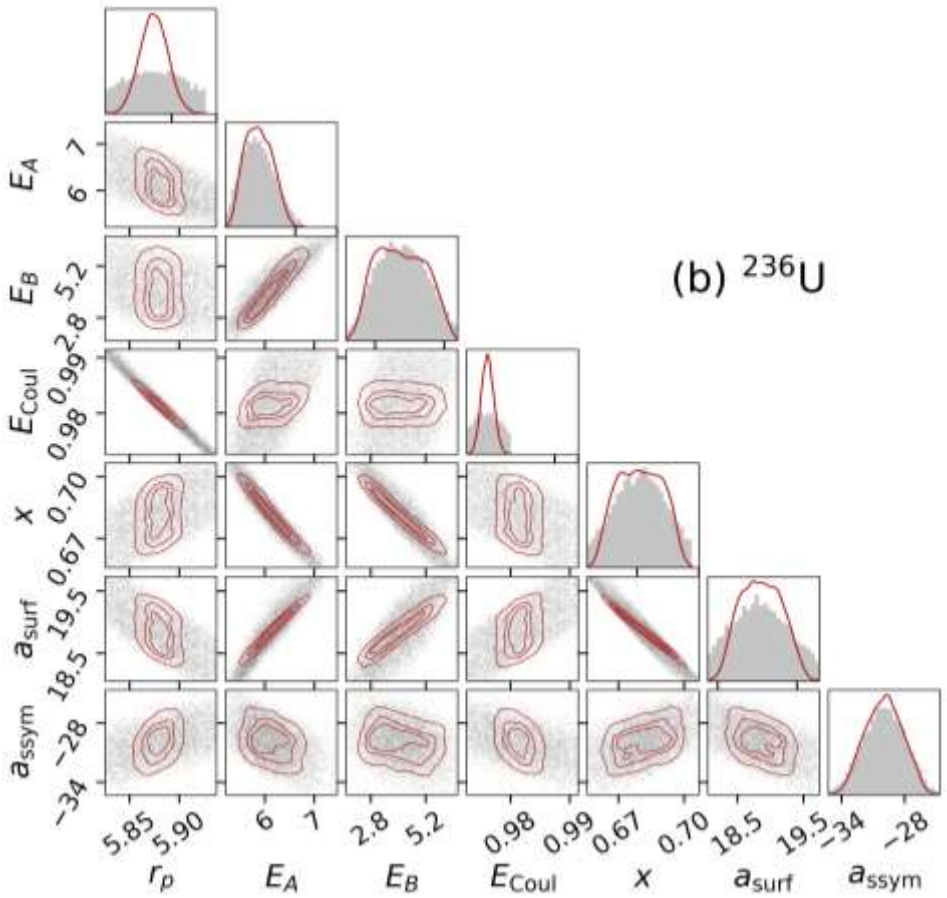
- The systematic error $\sigma_{\rm sys}$ for $O^{\rm pred}$ is estimated from the difference between the results computed using $N_f = 18$ and $N_f = 20$.

B.-N. Lu et al., Phys. Rev. C89, 014323 (2014).



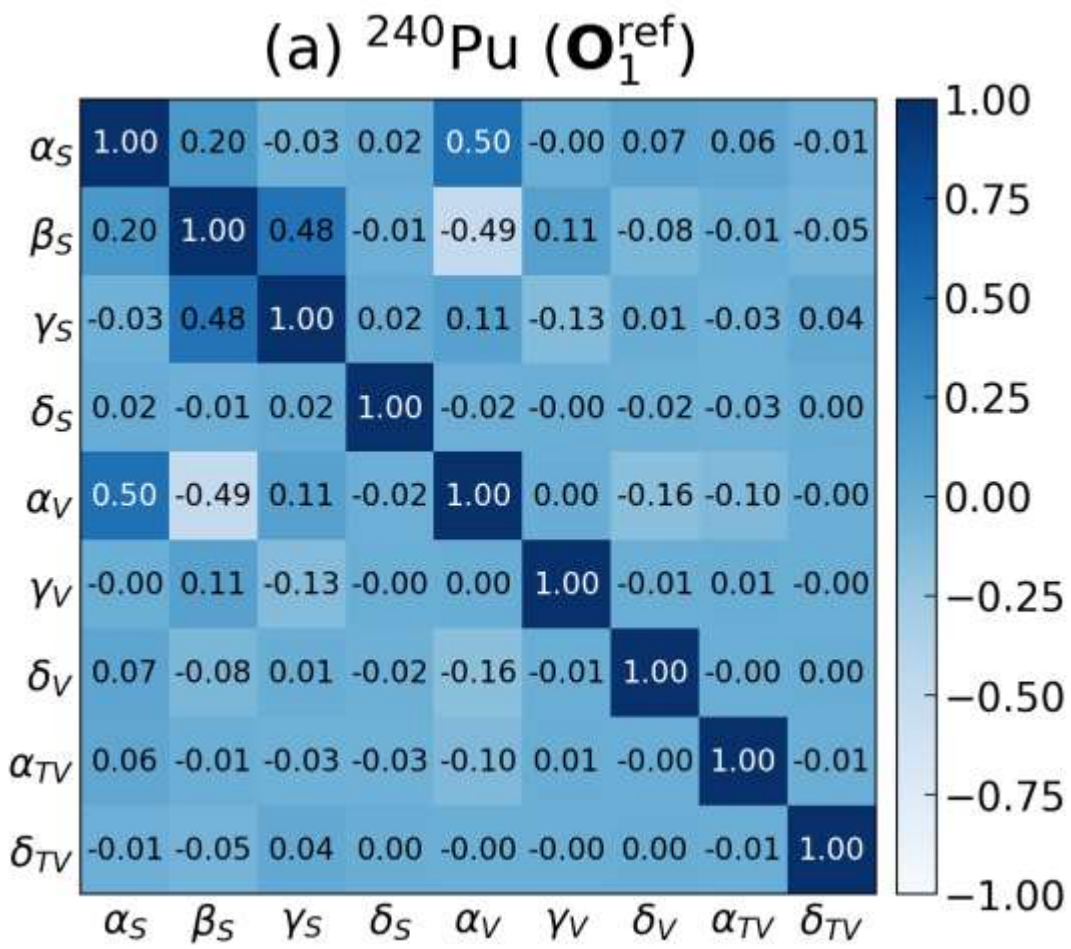
- $V(\beta_2) = E(\beta_2) - E_{\text{g.s.}}$
- The deformation of the ground state is highly stable.
- The uncertainty around the second fission barrier is larger than that around the first barrier.
- The equilibrium octupole deformation along the fission valley is insensitive to the EDF parameter set.

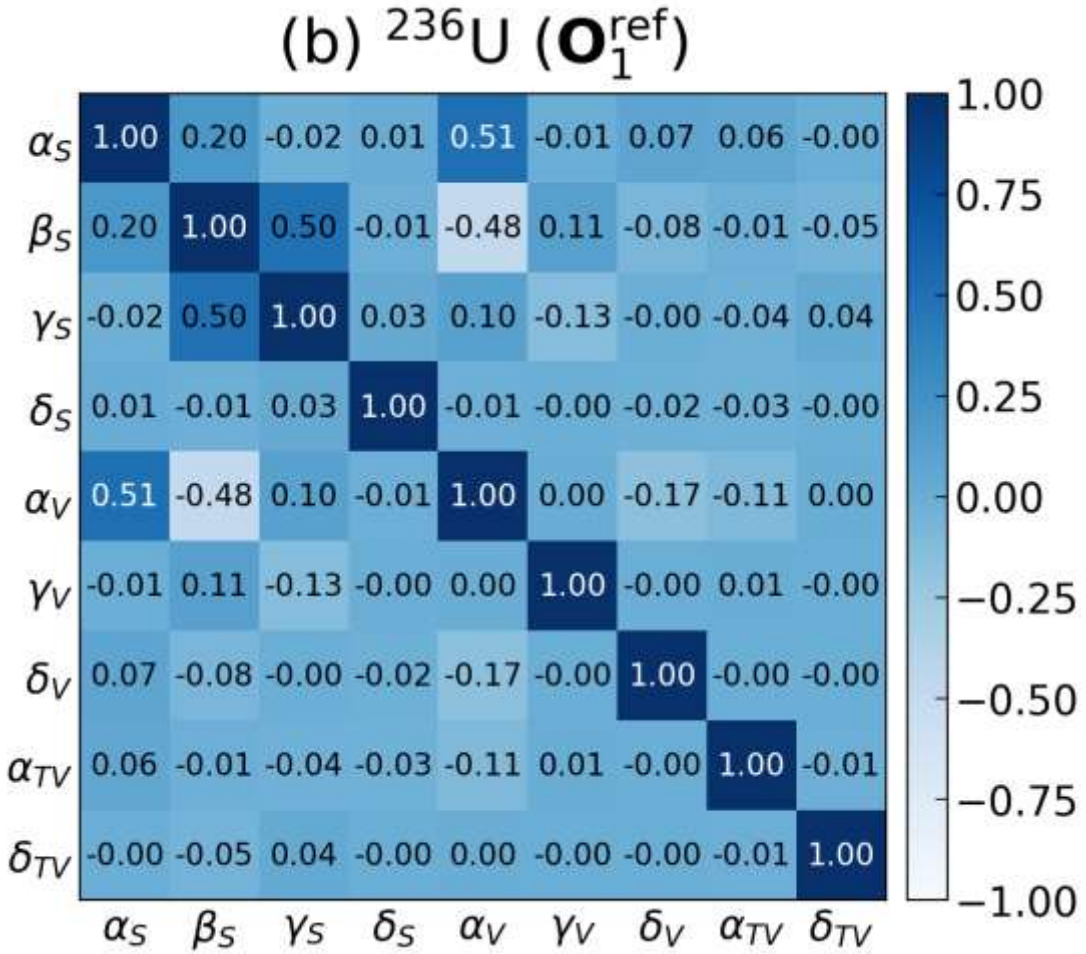


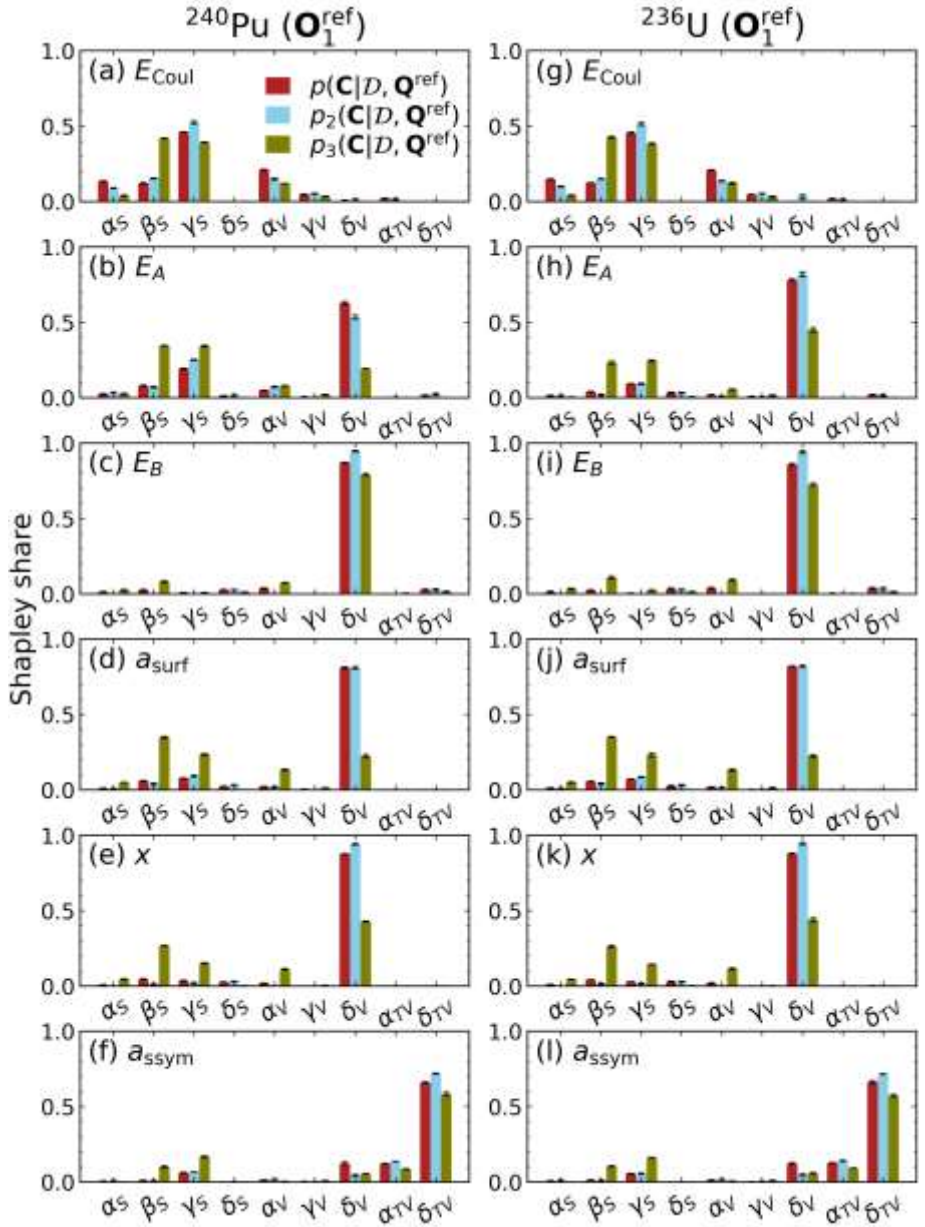


- The surface energy coefficient a_{surf} is positively correlated with both fission barrier E_A and E_B.

- The correlations between the low-energy constants from the posterior distributions $p(\mathbf{C}|\mathcal{D}, \mathbf{Q}^{\mathrm{ref}})$





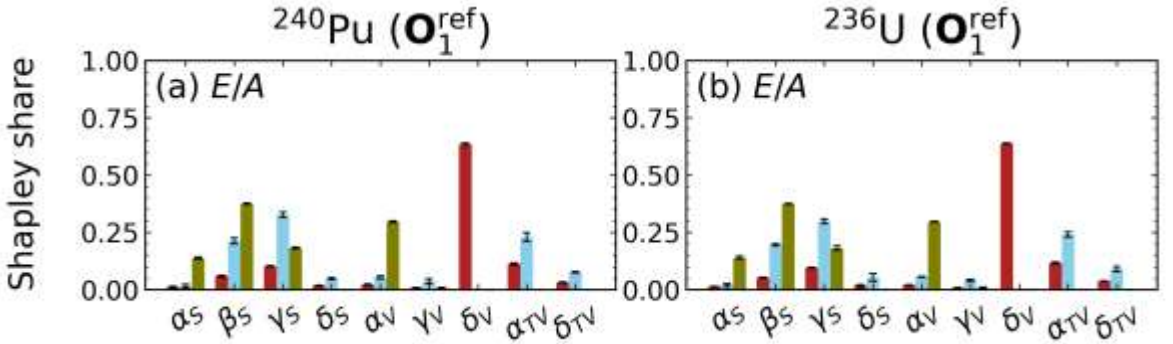


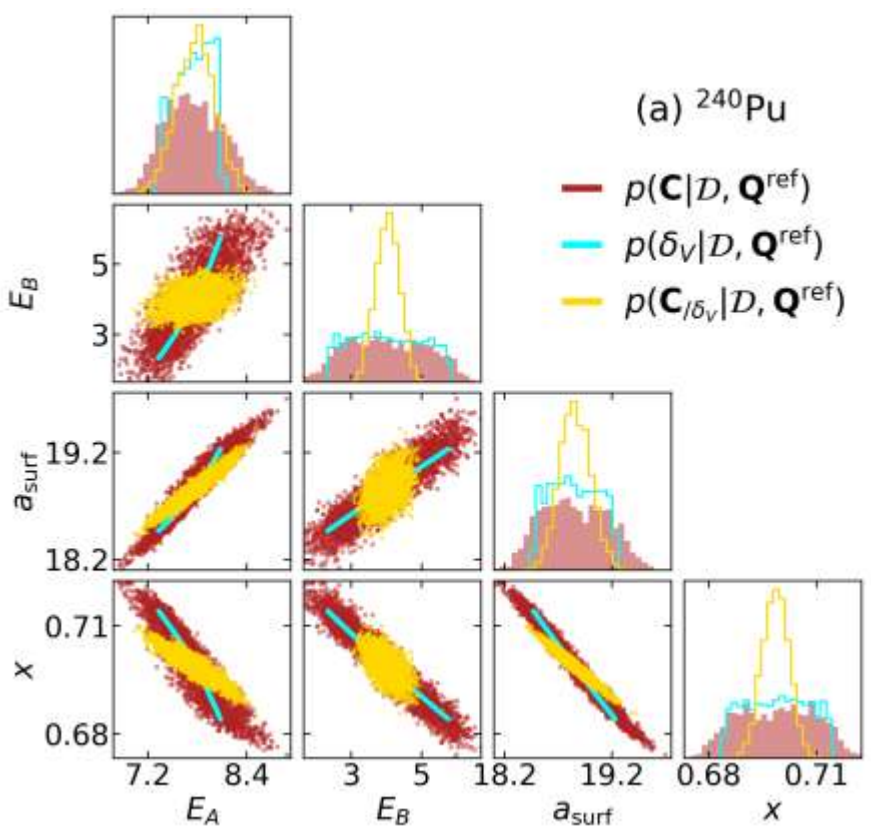
- Shapley global sensitivity analysis : The Shapley share index for parameter $C_i \in \mathbf{C}$ is defined as the average marginal contribution of C_i to the variance of the quantity O^{pred} over all possible subsets of parameters.

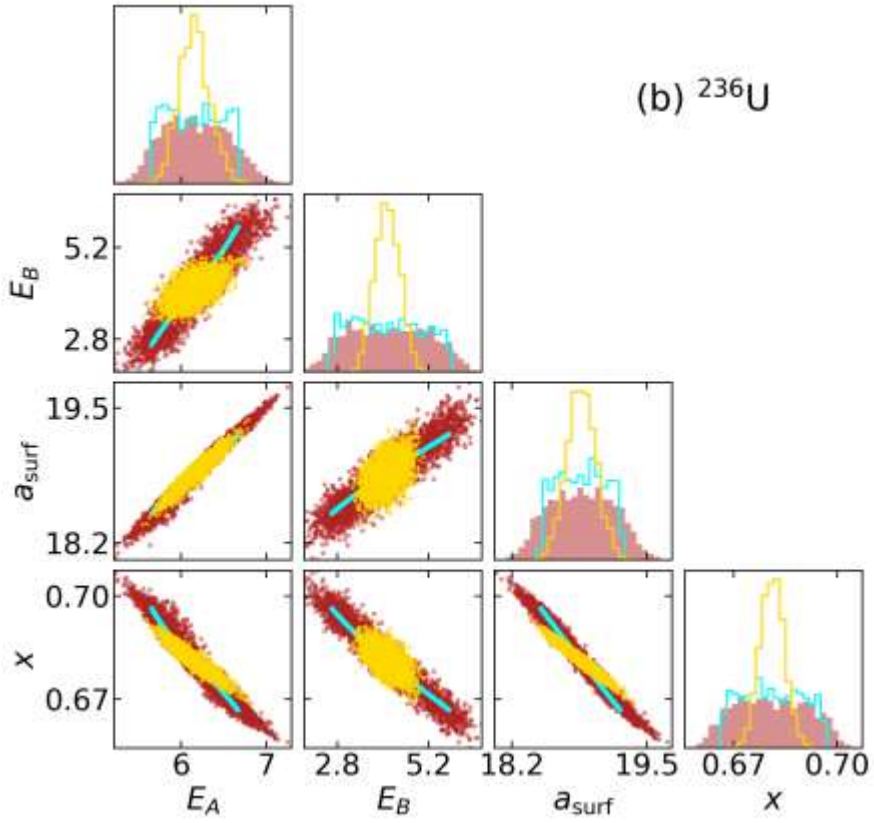
- Two factorized posterior distributions are used for comparison :

$$p_2(\mathbf{C}|\mathcal{D}, \mathbf{Q}^{\mathrm{ref}}) = p(\delta_V|\mathcal{D}, \mathbf{Q}^{\mathrm{ref}})\, p(\mathbf{C}_{/\delta_V}|\mathcal{D}, \mathbf{Q}^{\mathrm{ref}})$$

$$p_3(\mathbf{C}|\mathcal{D}, \mathbf{Q}^{\mathrm{ref}}) = \prod_{i=1}^{9} p(C_i|\mathcal{D}, \mathbf{Q}^{\mathrm{ref}})$$







- The outer barrier is predominantly sensitive to the isoscalar-vector derivative coupling, δ_V whereas the inner barrier and a_{surf} are jointly governed by δ_V, β_S, γ_S, and α_V, together with the correlations among these coupling constants.

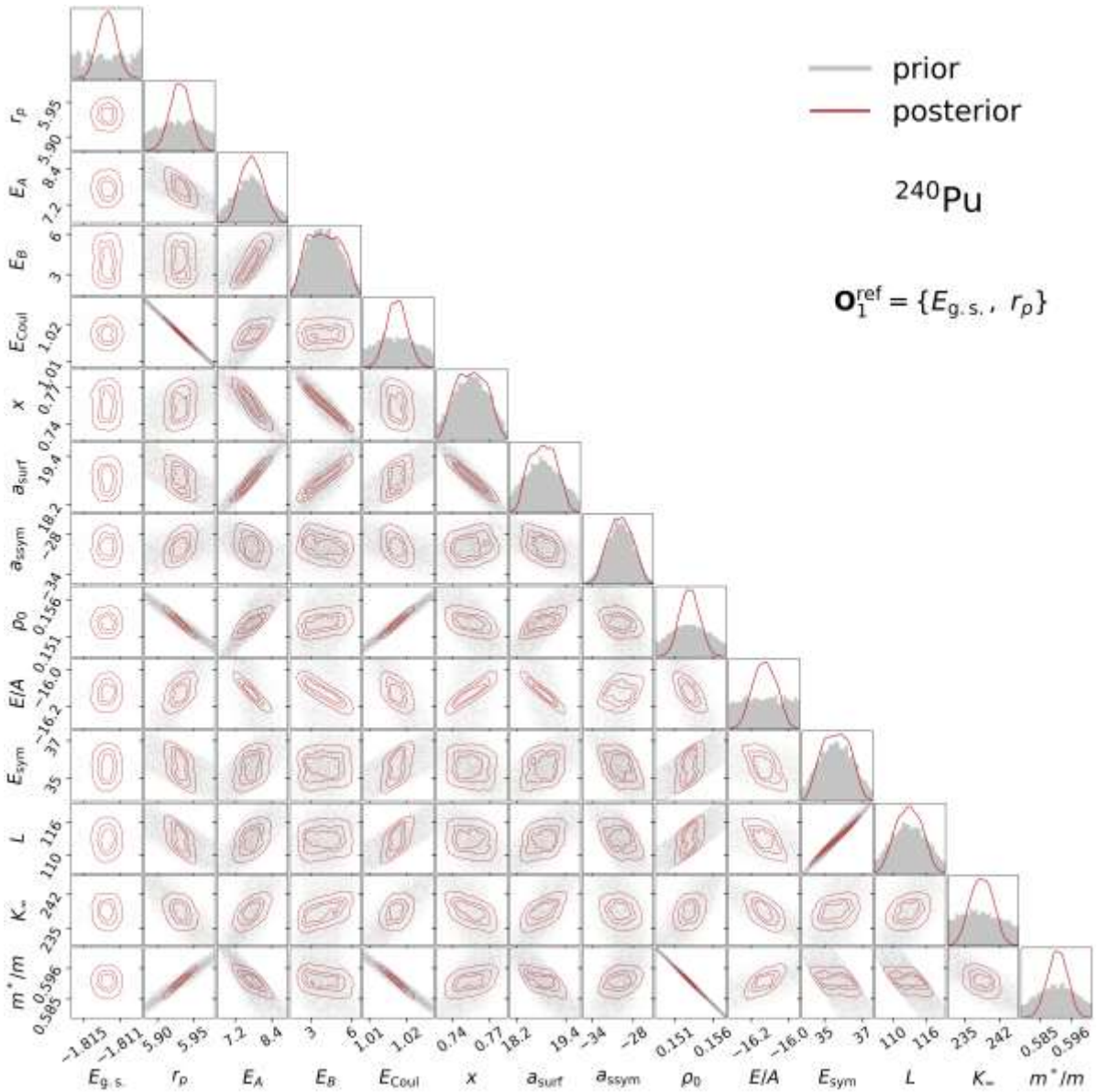
Summary

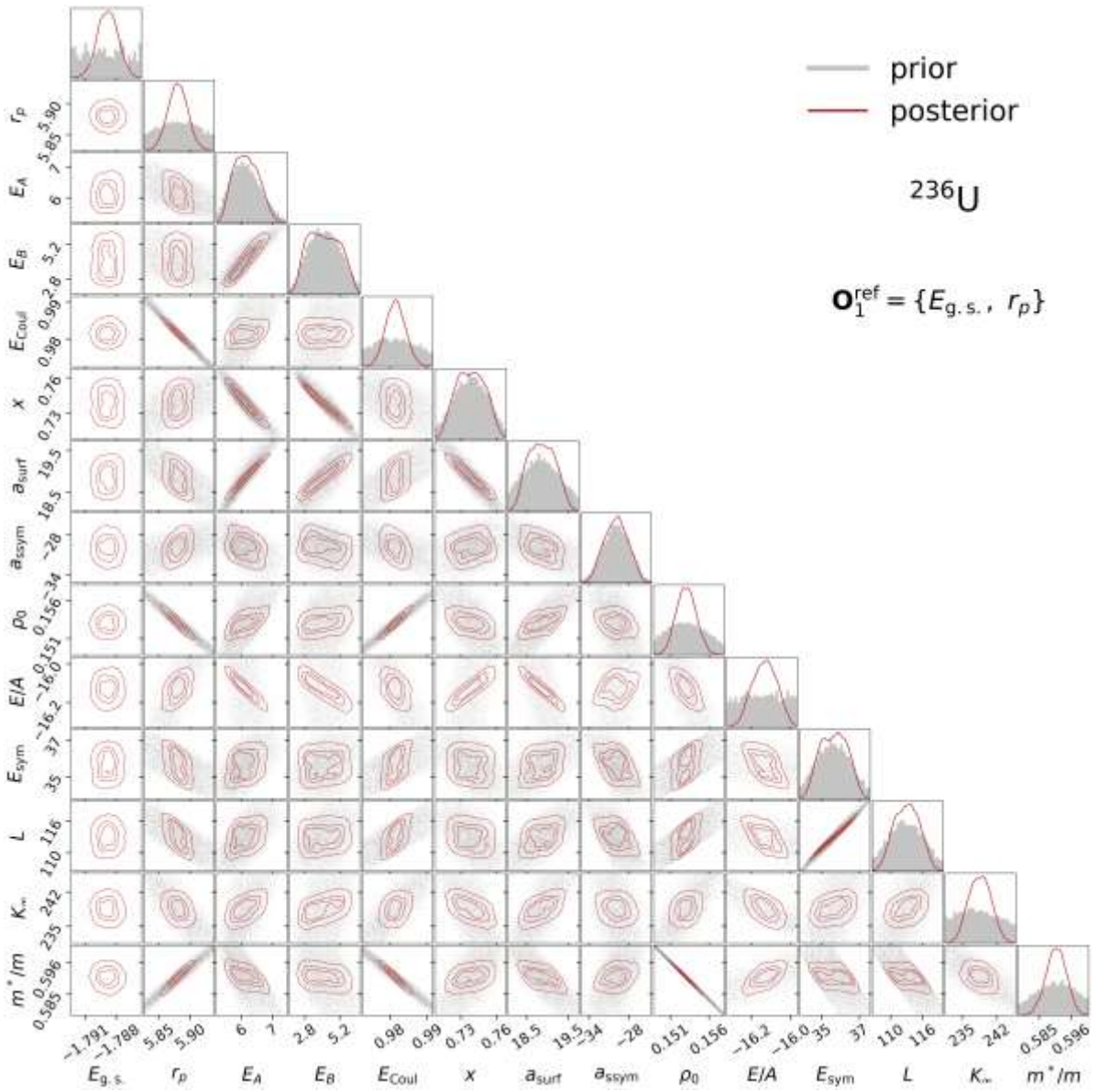
- We investigated the correlation between the surface energy coefficients and the fission properties of two representative actinide nuclei ^{240}Pu and ^{236}U.
- Results exhibit that the surface energy coefficient a_{surf} is strongly correlated with both fission barrier heights and the fissility parameter x, whereas the surface-symmetry energy coefficient a_{ssym} and the Coulomb energy E_{Coul} exhibit only weak correlations with the fission-related quantities.
- A sensitivity analysis further shows that the height of the outer fission barrier is governed predominantly by the isoscalar-vector derivative coupling constant δ_V whereas that of the inner barrier is by the combined effects of δ_V and several mutually correlated coupling constants in the EDF.

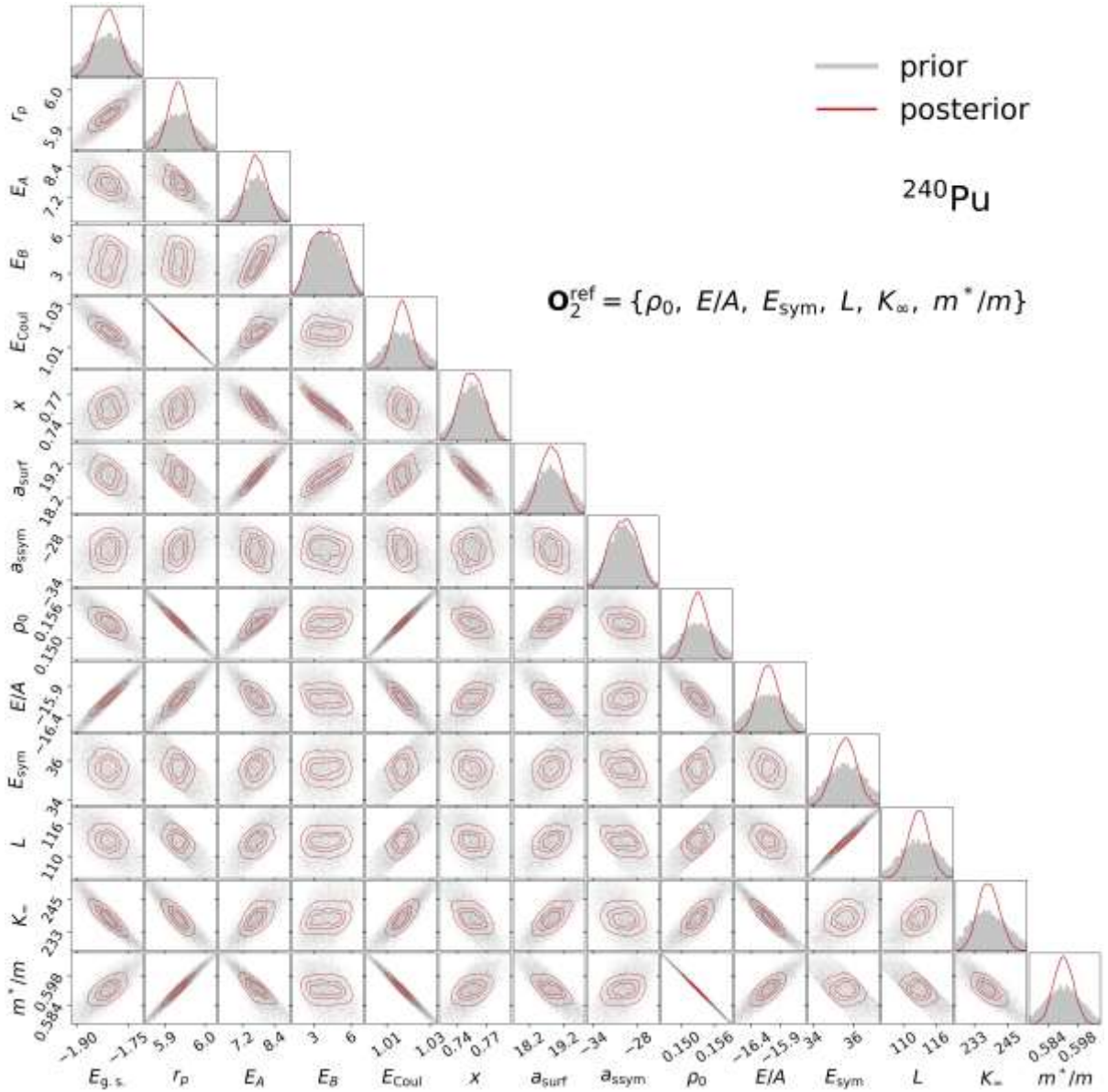
Prospect

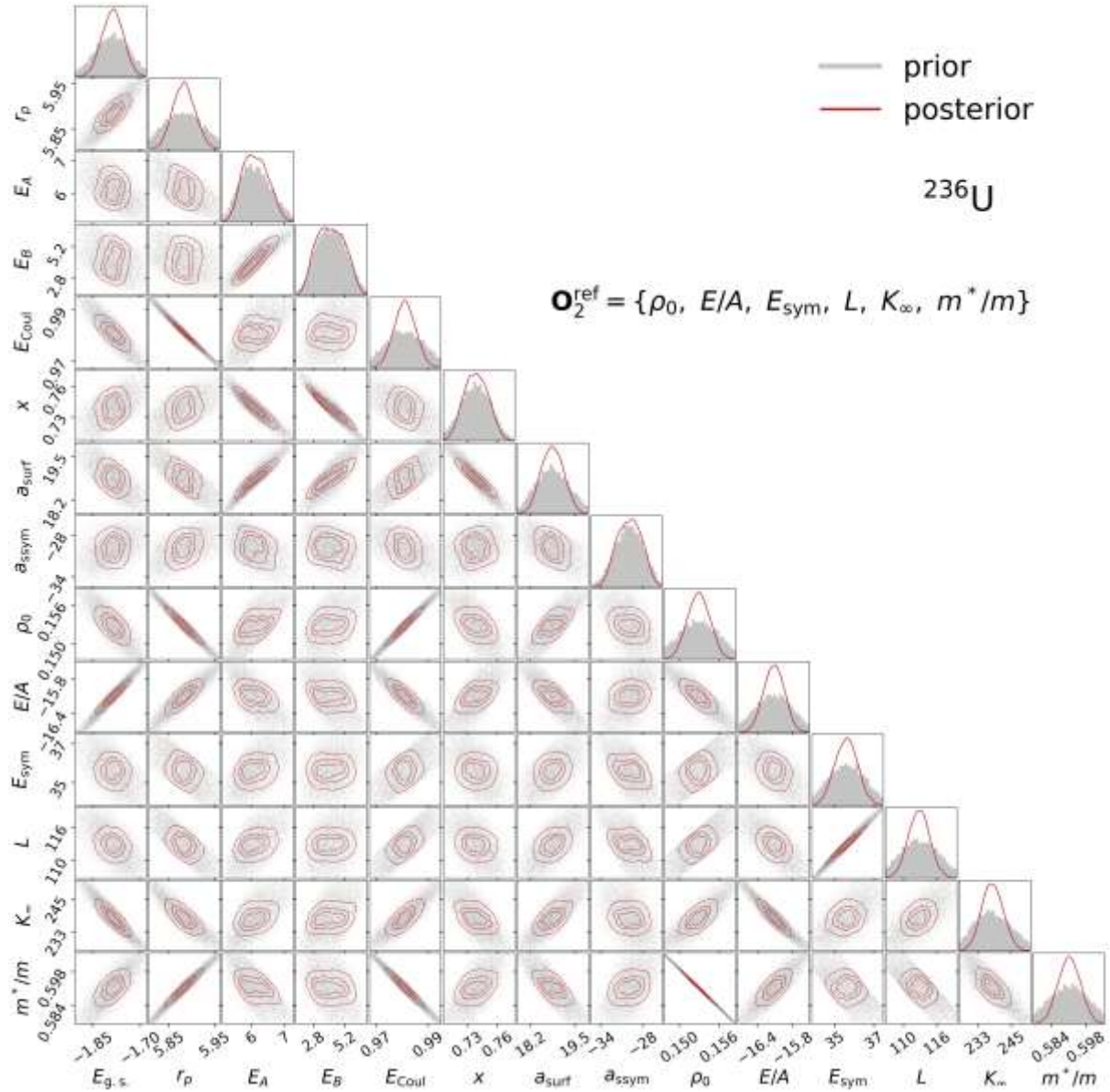
- We expect to extend this framework by including the triaxial deformation degree of freedom.
- Use fission data to constrain a_{surf} and low-energy constants ?

Thank you!


prior
posterior
^{240}Pu
$\mathbf{O}_1^{\mathrm{ref}} = \{E_{\mathrm{g.s.}},\ r_p\}$
$E_{\mathrm{g.s.}}$
r_p
E_A
E_B
E_{Coul}
x
a_{surf}
a_{ssym}
ρ_0
E/A
E_{sym}
L
K_∞
m^*/m


prior
posterior
^{236}U
$\mathbf{O}_1^{\mathrm{ref}} = \{E_{\mathrm{g.s.}},\ r_p\}$
$E_{\mathrm{g.s.}}$
r_p
E_A
E_B
E_{Coul}
x
a_{surf}
a_{ssym}
ρ_0
E/A
E_{sym}
L
K_∞
m^*/m


prior
posterior
^{240}Pu
$\mathbf{O}_2^{\mathrm{ref}} = \{\rho_0,\ E/A,\ E_{\mathrm{sym}},\ L,\ K_\infty,\ m^*/m\}$
$E_{\mathrm{g.s.}}$
r_p
E_A
E_B
E_{Coul}
x
a_{surf}
a_{ssym}
ρ_0
E/A
E_{sym}
L
K_∞
m^*/m


prior
posterior
^{236}U
$\mathbf{O}_2^{\mathrm{ref}} = \{\rho_0,\ E/A,\ E_{\mathrm{sym}},\ L,\ K_\infty,\ m^*/m\}$
$E_{\mathrm{g.s.}}$
r_p
E_A
E_B
E_{Coul}
x
a_{surf}
a_{ssym}
ρ_0
E/A
E_{sym}
L
K_∞
m^*/m